Hospital Medicine-Delivery Robot

Assignment 1 (CO1) - Output screenshots and graphs

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Task 8 - Correctness tests (console output)

```
$ python run_experiments.py   (Task 8: correctness tests)

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TASK 8 - CORRECTNESS TESTS
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Tiny graph (direct corridor 0-4 costs 10; detour 0-1-2-3-4 costs 4):
  BFS         path=[0, 4] hops=1 cost=10  valid=True
  UCS         path=[0, 1, 2, 3, 4] hops=4 cost=4  valid=True
  IDDFS       path=[0, 4] hops=1 cost=10  valid=True
  IDDFS-path  path=[0, 4] hops=1 cost=10  valid=True

1,000-state hospital (V=1002, undirected corridors=1498), graph seed=3026

  Test: reachable   start=0 goal=332
    BFS         found hops= 4 cost= 16 valid=True cost-optimal=True gen=364 exp=85
    UCS         found hops= 4 cost= 16 valid=True cost-optimal=True gen=1213 exp=318
    IDDFS       found hops= 4 cost= 16 valid=True cost-optimal=True gen=491 exp=109
    IDDFS-path  found hops= 4 cost= 16 valid=True cost-optimal=True gen=491 exp=109

  Test: start==goal   start=17 goal=17
    BFS         found hops= 0 cost=  0 valid=True cost-optimal=True gen=1 exp=0
    UCS         found hops= 0 cost=  0 valid=True cost-optimal=True gen=1 exp=0
    IDDFS       found hops= 0 cost=  0 valid=True cost-optimal=True gen=0 exp=0
    IDDFS-path  found hops= 0 cost=  0 valid=True cost-optimal=True gen=0 exp=0

  Test: unreachable (closed wing)   start=0 goal=1000
    BFS         NOT FOUND (correct=True)  gen=2995 exp=1000
    UCS         NOT FOUND (correct=True)  gen=2995 exp=1000
    IDDFS       NOT FOUND (correct=True)  gen=38225 exp=11634
    IDDFS-path  NOT FOUND (correct=True)  gen=3000001 exp=775150

All correctness assertions passed (found/not-found agree; BFS hops == IDDFS hops; UCS cost <= others).

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Task 9 - Performance, n = 1,000 and 10,000

```
$ python run_experiments.py   (Task 9: n = 1,000 and 10,000)

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TASK 9 - PERFORMANCE (start=0 pharmacy; 5 reachable wards per size; mean of 5 runs)
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n=    1,000  V=1,002  E(undirected)=1,498  preprocessing(build CSR adjacency)=0.00s  goals=[877, 33,
           820, 859, 609]
  BFS         found=5/5 capped=0 gen=       1,022 exp=       300 maxFrontier=      219 aux=      458
              depth=4.6 cost=25.8 optimal=0/5 valid=True time=0.000s
  UCS         found=5/5 capped=0 gen=       1,716 exp=       516 maxFrontier=      311 aux=      696
              depth=6.2 cost=20.6 optimal=5/5 valid=True time=0.001s
  IDDFS       found=5/5 capped=0 gen=       3,551 exp=       973 maxFrontier=        5 aux=      425
              depth=4.6 cost=25.8 optimal=1/5 valid=True time=0.002s
  IDDFS-path  found=5/5 capped=0 gen=       7,541 exp=     1,912 maxFrontier=        5 aux=        0
              depth=4.6 cost=25.8 optimal=1/5 valid=True time=0.004s

n=   10,000  V=10,002  E(undirected)=14,999  preprocessing(build CSR adjacency)=0.01s  goals=[4204,
           4720, 2390, 8907, 2809]
  BFS         found=5/5 capped=0 gen=      11,953 exp=     3,381 maxFrontier=    2,431 aux=    5,566
              depth=6.8 cost=34.4 optimal=0/5 valid=True time=0.006s
  UCS         found=5/5 capped=0 gen=      21,436 exp=     6,354 maxFrontier=    3,648 aux=    8,707
              depth=7.2 cost=30.2 optimal=5/5 valid=True time=0.028s
  IDDFS       found=5/5 capped=0 gen=      25,327 exp=     6,674 maxFrontier=        7 aux=    4,988
              depth=6.8 cost=34.4 optimal=2/5 valid=True time=0.018s
  IDDFS-path  found=5/5 capped=0 gen=      32,164 exp=     8,121 maxFrontier=        7 aux=        0
              depth=6.8 cost=34.4 optimal=2/5 valid=True time=0.020s
```

Task 9 - Performance, n = 100,000 and 1,000,000

```
$ python run_experiments.py   (Task 9: n = 100,000 and 1,000,000)

n=  100,000  V=100,002  E(undirected)=150,000  preprocessing(build CSR adjacency)=0.07s  goals=[13053,
           95482, 99249, 39227, 8647]
  BFS         found=5/5 capped=0 gen=     114,179 exp=    32,086 maxFrontier=   24,684 aux=   54,988
              depth=7.6 cost=44.4 optimal=0/5 valid=True time=0.090s
  UCS         found=5/5 capped=0 gen=     226,828 exp=    69,420 maxFrontier=   35,279 aux=   87,251
              depth=9.4 cost=41.6 optimal=5/5 valid=True time=0.486s
  IDDFS       found=5/5 capped=0 gen=     231,767 exp=    61,402 maxFrontier=        8 aux=   50,411
              depth=7.6 cost=44.4 optimal=1/5 valid=True time=0.169s
  IDDFS-path  found=5/5 capped=0 gen=     313,732 exp=    79,038 maxFrontier=        8 aux=        0
              depth=7.6 cost=44.4 optimal=1/5 valid=True time=0.236s

n=1,000,000  V=1,000,002  E(undirected)=1,499,998  preprocessing(build CSR adjacency)=0.99s
           goals=[812521, 282068, 223495, 337410, 710443]
  BFS         found=5/5 capped=0 gen=   1,138,289 exp=   322,535 maxFrontier=  237,679 aux=  538,355
              depth=9.2 cost=50.4 optimal=0/5 valid=True time=0.937s
  UCS         found=5/5 capped=0 gen=   2,472,786 exp=   764,618 maxFrontier=  360,102 aux=  926,201
              depth=10.4 cost=47.0 optimal=5/5 valid=True time=6.562s
  IDDFS       found=5/5 capped=0 gen=   2,527,127 exp=   671,545 maxFrontier=        9 aux=  478,972
              depth=9.2 cost=50.4 optimal=3/5 valid=True time=2.247s
  IDDFS-path  found=3/5 capped=2 gen=   1,739,676 exp=   436,625 maxFrontier=        9 aux=        0
              depth=8.3 cost=45.3 optimal=3/3 valid=True time=1.126s
```

Task 4 - Best / average / worst case

```
$ python run_experiments.py   (Task 4: best / average / worst case)

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TASK 4 - BEST / AVERAGE / WORST CASE (nodes generated)
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n=10,000: best goal=1 (depth 1), deepest reachable goal=9654 (depth 11), unreachable=10000
  BFS         best=         2  average=      10,277  deepest-reachable=      29,981  unreachable=      29,997
  UCS         best=        13  average=      18,902  deepest-reachable=      29,996  unreachable=      29,997
  IDDFS       best=         1  average=      19,554  deepest-reachable=     287,377  unreachable=     413,296
  IDDFS-path  best=         1  average=      23,778  deepest-reachable=   2,409,706  unreachable=   3,000,001 (CAPPED)

n=100,000: best goal=1 (depth 1), deepest reachable goal=74807 (depth 13), unreachable=100000
  BFS         best=         2  average=     110,482  deepest-reachable=     299,969  unreachable=     299,999
  UCS         best=        43  average=     183,491  deepest-reachable=     299,622  unreachable=     299,999
  IDDFS       best=         1  average=     230,606  deepest-reachable=   3,720,153  unreachable=   5,185,605
  IDDFS-path  best=         1  average=     296,539  deepest-reachable=   3,000,001  unreachable=   3,000,001 (CAPPED)

Saved results.json
```

Tasks 9-10 - Measured growth exponents

```
$ python make_plots.py   (growth exponents)

log-log slopes n:1e3->1e6
generated     BFS         slope=1.01  values=[1022, 11953, 114178, 1138289]
generated     UCS         slope=1.05  values=[1716, 21435, 226828, 2472786]
generated     IDDFS       slope=0.95  values=[3551, 25326, 231767, 2527126]
generated     IDDFS-path  slope=0.81  values=[7540, 32164, 313732, 1739675]
expanded      BFS         slope=1.01  values=[299, 3380, 32085, 322535]
expanded      UCS         slope=1.05  values=[516, 6353, 69420, 764618]
expanded      IDDFS       slope=0.95  values=[973, 6674, 61402, 671544]
expanded      IDDFS-path  slope=0.81  values=[1911, 8121, 79037, 436625]
max_frontier  BFS         slope=1.01  values=[219, 2431, 24684, 237679]
max_frontier  UCS         slope=1.02  values=[311, 3648, 35279, 360102]
max_frontier  IDDFS       slope=0.10  values=[4, 6, 7, 9]
max_frontier  IDDFS-path  slope=0.09  values=[4, 6, 7, 9]
aux_memory    BFS         slope=1.02  values=[457, 5566, 54987, 538355]
aux_memory    UCS         slope=1.04  values=[696, 8706, 87250, 926200]
aux_memory    IDDFS       slope=1.02  values=[424, 4988, 50411, 478972]
aux_memory    IDDFS-path  slope=0.00  values=[0, 0, 0, 0]
time          BFS         slope=1.13  values=[0.0, 0.006, 0.09, 0.937]
time          UCS         slope=1.25  values=[0.001, 0.028, 0.486, 6.562]
time          IDDFS       slope=1.03  values=[0.002, 0.018, 0.169, 2.247]
time          IDDFS-path  slope=0.84  values=[0.004, 0.02, 0.236, 1.126]
1000 prep 0.001 depth 4.6
10000 prep 0.007 depth 6.8
100000 prep 0.066 depth 7.6
1000000 prep 0.994 depth 9.2
```

Task 12 - Day-4 belief-state experiment

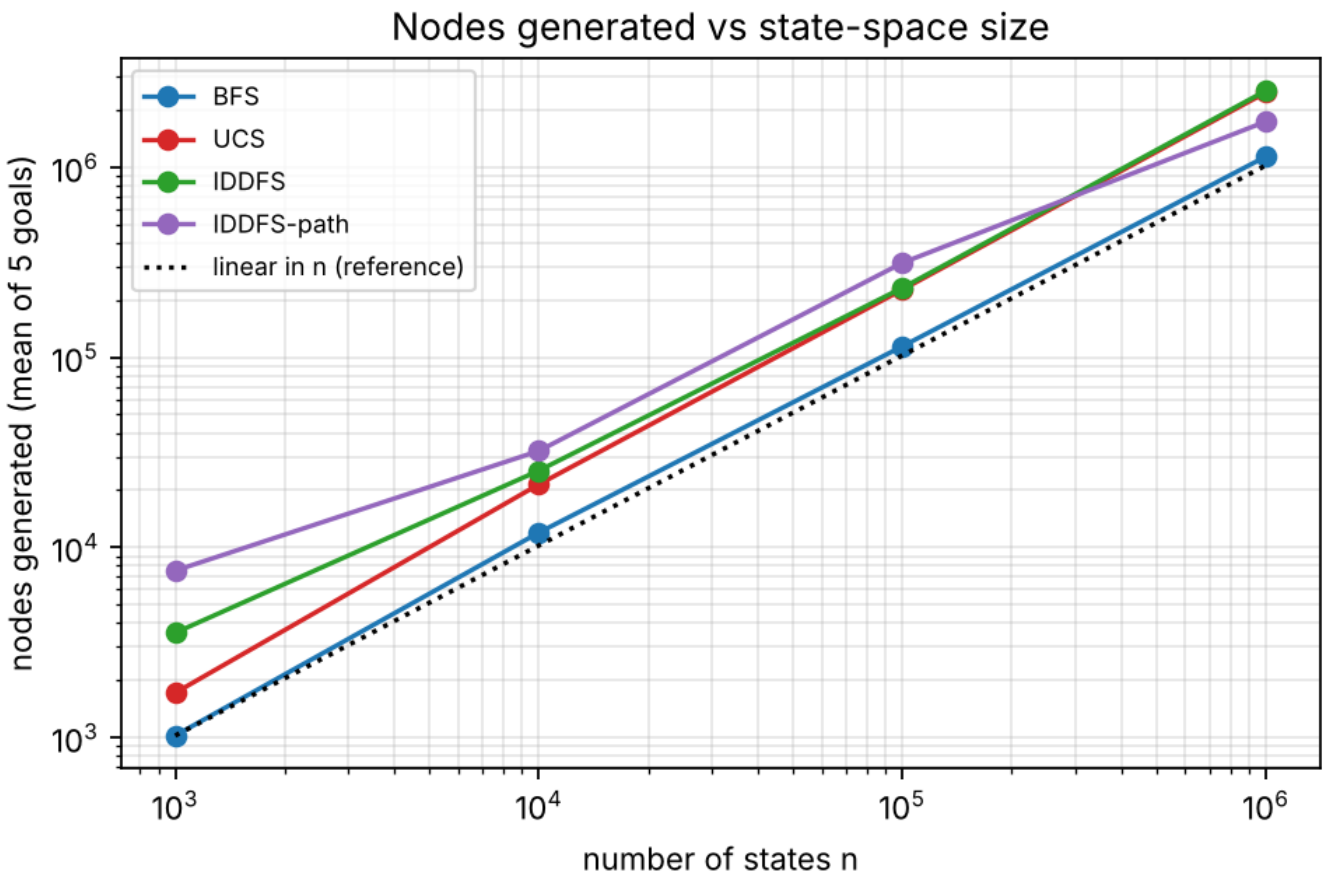
```
$ python belief_search.py   (Task 12 D)
TASK 12 D - Evidence (8x8 floor, 15% blocked cells, dock at (0,0), 8 random floors/belief sets per k)
Floor seeds 100..107; belief sets sampled with random.Random(1000*k+trial)

Columns: blind = success rate of the ORIGINAL plan from one guessed start;
         SL = sensorless belief-state BFS; SE = belief-state AND-OR search with local sensor
         solved = floors solved out of 8; gen = belief states generated; len = plan length (SE: worst
                  case)

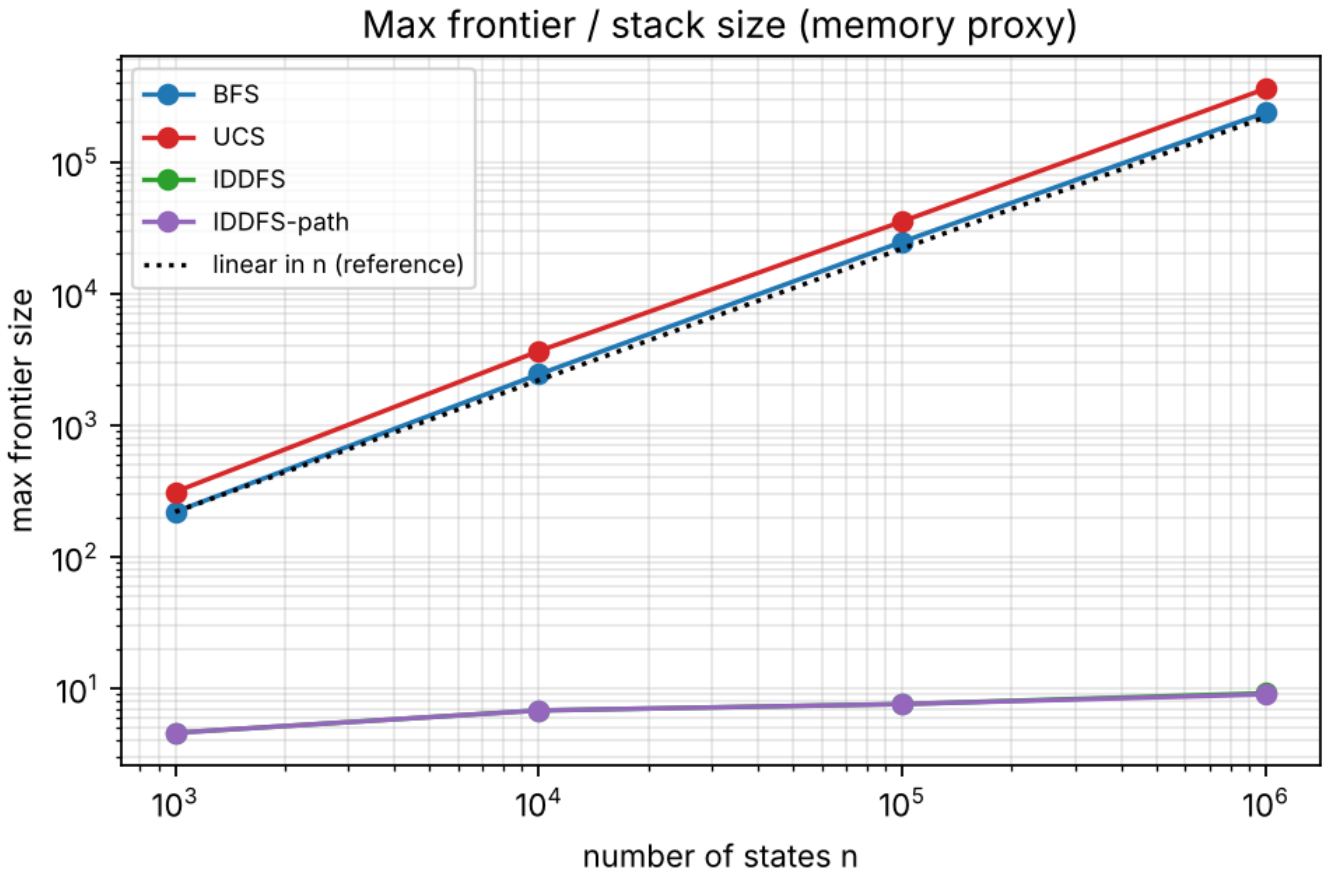
 k   blind | SL solved      SL gen  SL len | SE solved  SE gen  SE len
-------------------------------------------------------------------
 2   50.0% |       8/8        2319    10.1 |       8/8    47.4     9.5
 4   25.0% |       8/8      168454    12.6 |       8/8    52.4    11.5
 8   14.1% |       4/8     2243378    14.5 |       8/8    75.5    12.0

Size of the belief space (number of possible 'knowledge states'):
  n=       64: C(n,2)=2.016e+03  C(n,4)=6.354e+05  C(n,8)=4.426e+09   2^n-1 = 1.845e+19
  n=    1,000: C(n,2)=4.995e+05  C(n,4)=4.142e+10  C(n,8)=2.412e+19   2^n-1 = astronomical
  n=1,000,000: C(n,2)=5.000e+11  C(n,4)=4.167e+22  C(n,8)=2.480e+43   2^n-1 = astronomical
```

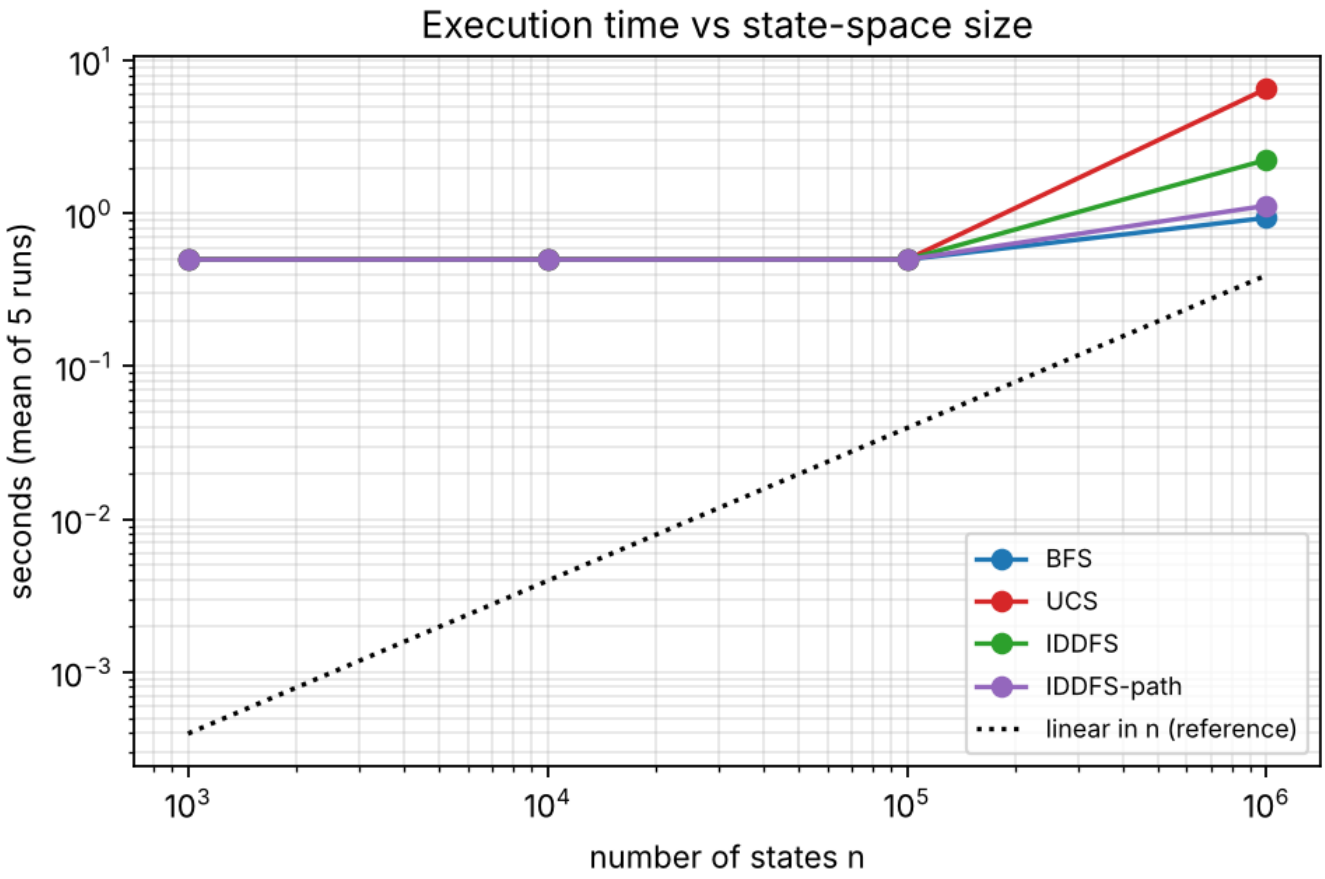
Graph - Nodes generated vs state-space size



Graph - Max frontier / stack size (memory proxy)



Graph - Execution time vs state-space size



Graph - Auxiliary memory (explored / depth-label entries)

